Farmer Name:	Ramesh Kumar	**Survey / Field ID:**	SF-248/B
Village & District:	Kammadikottai, Thoothukudi	**State:**	Tamil Nadu
Target Crop:	**Moringa**	**Crop Growth Stage:**	Flowering to Pod Initiation

75/100	**Suitable**	**ACTIVE**	**7-DAY**
SOIL HEALTH INDEX	CROP COMPATIBILITY	BIOBOOST STATUS	AGROMET FORECAST
Rating: Good	Target: Moringa	Microbial & Organic	Synced with Open-Meteo

1. SOIL LAB & FERTILITY INTELLIGENCE DASHBOARD

Physico-Chemical & Nutrient Status

Soil Health Index: 75/100 (Good)

Benchmark: *ICAR Soil Health Card Scheme standard nutrient rating thresholds.*

Soil Parameter	Measured Value	Status / Rating	Standard Agronomic Benchmark
pH (Soil Acidity/Alkalinity)	6.8	**OPTIMAL**	6.5 - 7.5 (Optimal Range)
Available Nitrogen (N)	285.0 kg/ha	**OPTIMAL**	280 - 560 kg/ha (Medium / Desirable)
Available Phosphorus (P)	18.4 kg/ha	**OPTIMAL**	11 - 25 kg/ha (Medium / Desirable)
Available Potassium (K)	210.0 kg/ha	**OPTIMAL**	140 - 280 kg/ha (Medium / Desirable)
Organic Carbon (OC)	0.62 %	**OPTIMAL**	> 0.75 % (High / Optimal Biological Reserve)
Electrical Conductivity (EC)	0.35 dS/m	**OPTIMAL**	< 1.0 dS/m (Non-saline / Safe Emergence)

Key Soil Strengths:	Balanced soil pH and favorable non-saline conditions.
Nutrient Concerns:	Maintain organic matter and routine fertility replenishment.

2. CROP SUITABILITY & AGRONOMIC PROFILE

Soil-Crop Compatibility & Phenology

Crop Intelligence Unavailable	*Crop data is Not available. Field and crop intelligence, soil suitability evaluation, phenological growth stages, and tailored cultivation guidelines will activate once a crop profile is selected.*

3. BIOBOOST / BIOLOGICAL NUTRIENT MANAGEMENT

Soil Biology & Organic Fertility Framework

SOIL FINDING & NUTRIENT STATUS

• **Organic Carbon Reserve:** Current level is **0.62%**. Soil organic matter is the primary biological substrate required to sustain beneficial rhizosphere microorganisms.
• **Macronutrient Baseline:** Nitrogen: 285.0 kg/ha • Phosphorus: 18.4 kg/ha • Potassium: 210.0 kg/ha. Biological solubilizers are recommended to enhance uptake efficiency and minimize chemical fixation in soil.

BIOLOGICAL RECOMMENDATIONS (BIOFERTILIZERS & MICROBIALS)

• **Nitrogen Fixation:** Inoculate with *Azospirillum* or *Rhizobium* to fix atmospheric nitrogen and stimulate root elongation.
• **Phosphorus Solubilizers:** Apply *Phosphobacteria* (PSB) to mobilize insoluble calcium/aluminum phosphates into plant-available orthophosphates.
• **Potassium Mobilizers:** Apply *Frateuria aurantia* (KMB) to mobilize mineral potassium from silicate lattices.
• **Mycorrhizal Fungi (VAM):** Incorporate *Glomus* species to extend root surface area for enhanced phosphorus and water uptake.
• **Inoculation Protocol:** Mix biofertilizers with well-decomposed FYM or vermicompost in a shaded area 24 hours prior to field application.

ORGANIC MATTER & SOIL HEALTH MANAGEMENT

• **Farmyard Manure (FYM):** Incorporate well-rotted FYM or compost during final land preparation to rebuild microbial population and enhance cation exchange capacity.
• **Vermicompost Application:** Top-dress vermicompost around active feeder roots to supply bio-available humic substances and micronutrients.
• **Green Manuring:** Sow and incorporate green manure crops (e.g. *Sunnhemp* or *Dhaincha*) at 50% flowering to add 15–20 tonnes of green biomass per hectare.
• **Organic Mulching:** Maintain a 5–7 cm layer of dry crop residues or leaf mulch across planting rows to reduce evaporative water loss and moderate soil temperature.

AGRONOMIC VERIFICATION & DOSAGE SAFETY POLICY

Mandatory Stewardship Notice: AgriAI strictly adheres to responsible agronomic safety standards and does **NOT** fabricate commercial synthetic fertilizer dosages or arbitrary product quantities. Commercial fertilizer inputs must be calibrated strictly against your local **Soil Testing Laboratory (STL) prescription** and official **State Agricultural University (TNAU / ICAR / KVK) Package of Practices**. Always consult your Block Agricultural Extension Officer before purchasing chemical fertilizers.

4. INTEGRATED PEST MANAGEMENT (IPM) DOSSIER

Ecological Decision Support & Diagnostic Guidance

Target Pest / Disease:	**Moringa Fruit Fly (Gitona distigma) (Insect Pest)**	**Risk Severity:**	**LOW RISK • PREVENTIVE**
Visual Symptoms:	Internal feeding by maggots causes premature pod drying, splitting, and gummy exudation on pod surface.	**Monitoring Protocol:**	Install methyl eugenol / yellow sticky traps (5-10 traps/acre). Inspect 20 plants across 5 field spots weekly.

HIERARCHICAL IPM INTERVENTION PROTOCOL

• **Cultural & Mechanical Control:** Collect and destroy fallen or infested pods; deep summer ploughing to expose puparia to natural solar radiation and predators.
• **Biological & Biorational Control:** Conserve parasitoids (Opius sp.); spray 5% Neem Seed Kernel Extract (NSKE) or Azadirachtin 10,000 ppm at initial flower bud formation.
• **Ecological Threshold Level (ETL):** Initiate curative control measures only when trap catches or field sampling confirm that pest density exceeds the Economic Threshold Level.
• **Pollinator Conservation:** Avoid all field sprays during peak morning hours (06:00 AM – 09:30 AM) when honeybees and pollinators are actively foraging on blossoms.

STATUTORY CHEMICAL CONTROL & PESTICIDE DOSAGE SAFETY

Regulatory Label Compliance Notice:
• **Zero Fabricated Dosages:** AgriAI strictly prohibits generating arbitrary pesticide doses, tank mixes, or unverified spray schedules.
• **Approved Active Ingredients Only:** Chemical pesticides must be selected exclusively from the registered list approved by the **Central Insecticides Board & Registration Committee (CIBRC)** for the specific target crop and pest.
• **Container Label Mandate:** Follow the current registered product label and local agricultural extension recommendation for dosage, water dilution, personal protective equipment (PPE), and mandatory Pre-Harvest Intervals (PHI).

5. WEATHER & 7-DAY AGROMET FORECAST

Live Microclimate & Operational Farm Windows

31.2 °C	68 %	12.5 km/h	0.0 mm
TEMPERATURE	HUMIDITY	WIND SPEED	PRECIPITATION
Current Ambient	Relative Humidity	Gentle Breeze	Last 24 Hours

Date / Day	Temp (Min-Max)	Rain Probability	Rain (mm)	Wind Speed	Agromet Operational Advisory
Day +1	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +2	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +3	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +4	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +5	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +6	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.
Day +7	24° - 32°C	**15%**	0.0 mm	11 km/h	Favorable conditions for routine intercultural operations.

FARM AGROMET OPERATIONAL WINDOWS

• **Foliar Spraying Window:** Favorable during early morning hours (06:30 AM – 09:00 AM) when wind speed is below 15 km/h and rain probability is under 30%. Never spray during high thermal inversion or gusty winds.
• **Irrigation Scheduling:** Soil moisture reserves should be verified prior to scheduling supplementary irrigation. Delay irrigation if 7-day forecast indicates precipitation exceeding 15 mm.
• **Field Trafficability:** Current dry-to-moderate surface conditions ensure safe tractor traffic and manual weeding without causing soil compaction or smear.

Agrometeorological forecast data provided by Open-Meteo High-Resolution Numerical Weather Prediction (NWP) API. Local microclimates may vary with field elevation.

6. OFFICIAL IMD MONSOON OUTLOOK

National Seasonal Meteorological Assessment

Latest verified IMD outlook unavailable

Official India Meteorological Department (IMD) seasonal bulletin records could not be verified from the server at this time. AgriAI adheres strictly to factual integrity and **never presents stale historical meteorological information as current outlook**.

Farmer Recommendation: Consult your local **Agromet Field Unit (AMFU)**, District Agromet Bulletin (GKMS), or the official IMD Mausam portal directly at https://mausam.imd.gov.in/ for current seasonal and medium-range weather bulletins.

7. FARMER ACTION PLAN & MONITORING PROTOCOL

Prioritized Interventions & Weekly Milestones

PRIORITY 1: IMMEDIATE INTERVENTION (DAYS 1–7)

• **Soil & Seedbed Preparation:** Perform basal land preparation; incorporate well-decomposed FYM or compost to address soil structure and fertility balance.

PRIORITY 2: BIOLOGICAL INPUTS & SEED CARE (DAYS 8–21)

• **Rhizosphere Microbial Priming:** Inoculate planting seed or seedling roots with target biofertilizers (Azospirillum & Phosphobacteria) to foster root branching in Moringa.

PRIORITY 3: AGROMET SYNCHRONIZATION & IPM SCOUTING (ONGOING)

• **Field Monitoring & Water Scheduling:** Synchronize field operations with the 7-day weather forecast. Install yellow sticky / pheromone traps for pest scouting.

Inspection Window	Target Assessment Item	Diagnostic Threshold / Action Trigger	Verification Sign-off
Days 1 – 3	Soil Moisture & Tilth	Inspect 15-30 cm root zone; confirm soil moisture before sowing.	[ ] Verified
Days 4 – 7	Seedling Emergence	Examine germination count; verify no damping-off or water stagnation.	[ ] Verified
Days 8 – 14	Pest Scouting & Traps	Inspect lower leaf surfaces; count trap catches against ETL.	[ ] Verified
Days 15 – 21	Canopy Vigor & Roots	Confirm biofertilizer response and active vegetative vigor.	[ ] Verified
Days 22 – 30	Intercultural Weeding	Perform shallow weeding or mulching to suppress weed competition.	[ ] Verified

EXTENSION & REGULATORY VERIFICATION CHECKLIST

• [ ] **Soil Test Prescription:** Confirm commercial fertilizer requirements with your local Soil Testing Laboratory (STL) prescription slip.
• [ ] **Extension Consultation:** Cross-verify crop variety suitability with the Block Agricultural Extension Officer (AEO).
• [ ] **Input Authenticity:** Source certified seeds and authorized biofertilizers from government agricultural depots or university counters.
• [ ] **Label Directions:** Check CIBRC registration and container labels before applying any plant protection product.

8. TECHNICAL SPECIFICATIONS & REGULATORY PROVENANCE

Data Integrity, Source Attribution & Disclaimer

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Compilation Timestamp	25 Sep 2026, 08:27 PM
Farmer & Field Identity	Ramesh Kumar • Survey No: SF-248/B • Kammadikottai, Thoothukudi, Tamil Nadu
Agricultural Intelligence Platform	AGRIORA (AgriAI) Smart Farming Decision-Support System v2.0
Decision Inference Engine	Deterministic, transparent rule-based agronomic algorithms (Zero generative hallucinations)

AUTHORITATIVE SCIENTIFIC & INSTITUTIONAL DATA SOURCES

- **Indian Council of Agricultural Research (ICAR):** Crop agronomy, physiological benchmarks, and crop-specific management standards.
- **Tamil Nadu Agricultural University (TNAU):** Agritech Portal crop production packages, soil fertility thresholds, and local IPM guides.
- **India Meteorological Department (IMD):** Official national seasonal monsoon bulletins, regional forecasts, and agromet advisories under Ministry of Earth Sciences, Govt. of India.
- **Open-Meteo Weather API:** High-resolution global numerical weather prediction (NWP) model forecasts.
- **Central Insecticides Board & Registration Committee (CIBRC):** Approved pest management chemistries, safety guidelines, and pesticide regulations under Ministry of Agriculture.
- **Soil Health Card Scheme:** Standard nutrient classification ratings (Ministry of Agriculture & Farmers Welfare).

STATUTORY AGRONOMIC DISCLAIMER

AGRIORA provides advisory decision support based on farmer-supplied soil test records, verified meteorological bulletins, and established institutional agronomic benchmarks. While every analytical model is calibrated against authoritative university and governmental standards, this report is designed solely for field-level planning. It does **NOT** constitute a legal warranty of crop yield, financial guarantee, or substitute for mandatory on-site inspection by a qualified agricultural scientist, Krishi Vigyan Kendra (KVK) expert, or government extension agent. Commercial input applications (fertilizers, pesticides, amendments) must comply strictly with statutory regulations, registered product labels, and localized extension recommendations.

AGRIORA — SCIENCE-DRIVEN FARM PROSPERITY

Dedicated to empowering Indian farmers through transparent agronomic intelligence, data integrity, and sustainable soil stewardship.